



Faculty Of Engineering and Geology.

Computer and Data Communication

**Digital Communication System Implementation
(Manchester Encoding/Decoding)**

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- **Abstract:**

This project presents a complete digital communication system implemented in LabVIEW NXG using Manchester line encoding. The system reads data, converts it into a Manchester-encoded waveform, transmits it through a simulated channel, and reconstructs the original data using a precise decoding process. The results show that the receiver successfully recovers the transmitted bits, reconstructing the exact original data with a zero bit-error rate, demonstrating the reliability of Manchester encoding for synchronization and data integrity.

- **Introduction and Theoretical Background:**

Manchester encoding is one of the fundamental line-coding techniques used in digital communication systems to ensure reliable data transmission. In this encoding scheme, the logic state of a bit is determined by the direction of a voltage transition occurring exactly in the middle of the bit period. Typically, a logic '0' is represented by a transition from a **High to a Low voltage level**, while a logic '1' is represented by a transition from a **Low to a High voltage level**. This continuous mid-bit transition acts as a built-in clock signal, allowing the receiver to perfectly synchronize with the transmitter without requiring an external clock, even during long sequences of consecutive ones or zeros.

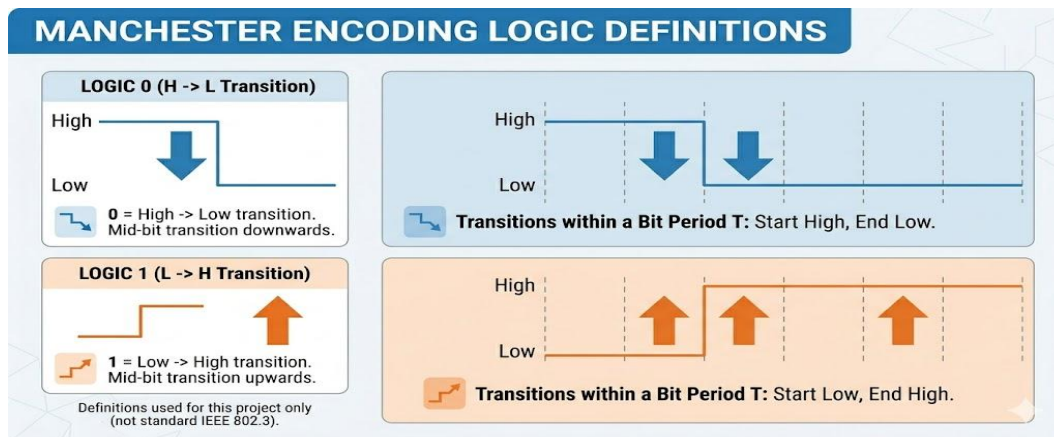


Figure 1: Generated Manchester encoding principle.

Building on this principle, this project simulates a complete sender-to-receiver communication model using LabVIEW NXG. The system reads input binary data, encodes it into a Manchester waveform, transmits the signal, and finally decodes it back into the original data. LabVIEW NXG provides an ideal environment for this implementation through visual programming and real-time waveform visualization. The project highlights essential engineering concepts such as bit mapping, decoding logic, array decimation, and critical bit-order correction (MSB/LSB alignment) to accurately reconstruct the digital signal representation.

• Experimental Procedure:

This section details the methodical implementation of the Manchester Transceiver system within **LabVIEW NXG**. The procedure is categorized into three critical phases: Data Input/Conversion (Reading), Encoding & Display, and Decoding & Verification.

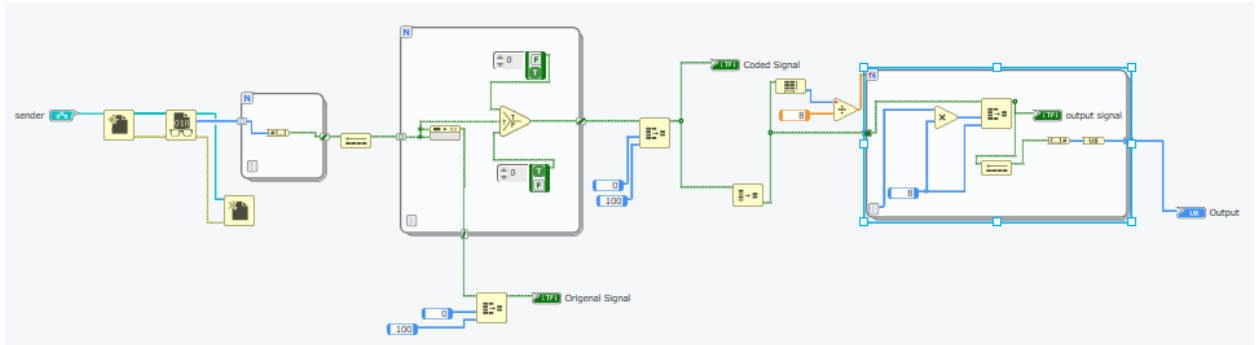


Figure 2: Complete Block Diagram of the Manchester Transceiver system in LabVIEW NXG.

Phase 1: Data Input and Binary Conversion

Initially, the system was configured to read textual input data from the user. A String Control, labeled sender, accepts the input character (e.g., 'A'). This character is dynamically converted into its constituent byte values and subsequently transformed into an 8-bit **Boolean Array** representation to represent the raw binary data.

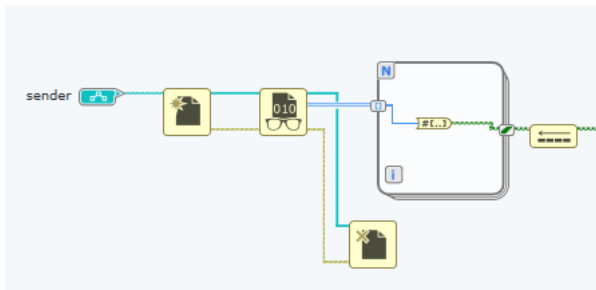


Figure 3: Phase 1: Data Input (String) and conversion to 8-bit Boolean Array

Phase 2: Manchester Encoding and Waveform Display

The raw Boolean array is processed inside a For Loop to apply the Manchester encoding logic. Based on the established logic rules for this project (0: High-to-Low transition, 1: Low-to-High transition), each original bit is mapped to a doubled-bit representation to embed the clock transition in the mid-bit period. The resulting expanded array forms the **Encoded Signal**. Both the **Original Signal** and the **Coded Signal** waveforms are fed simultaneously into Waveform Charts to visualize the real-time waveforms and verify the required transitions.

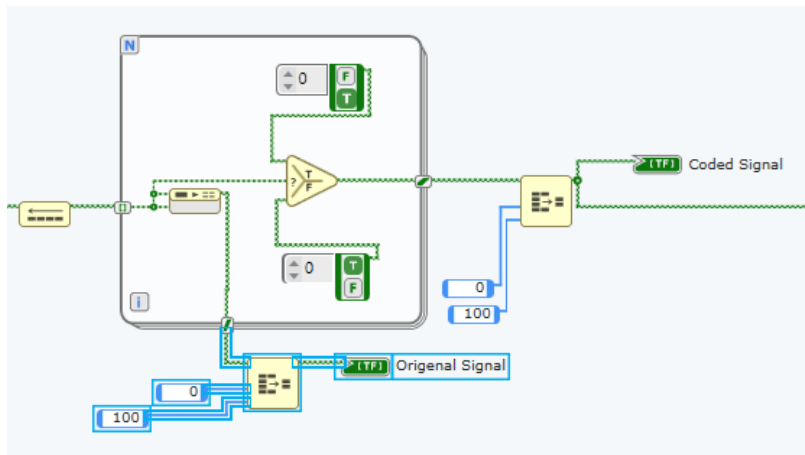


Figure 4: block diagram of Phase 2 encoding logic

Phase 3: Manchester Decoding and Final Verification

This phase demonstrates the accurate recovery of the original data at the receiver. The decoding process involves decimating the encoded Manchester stream (e.g., extracting odd-indexed bits) to reconstruct the single representative data bits from the doubled stream.

The Critical Verification Step (Bit Reversal): To correct for LabVIEW NXG's default bit significance order (which standardly processes arrays from the least significant bit **LSB** first), a **Reverse 1D Array** function was applied. This step ensures standard **MSB (Most Significant Bit)** alignment required for standard ASCII value calculation.

The reversed, decimated Boolean array is then converted into an 8-bit Unsigned Integer (U8) numeric value and displayed on a Numeric Indicator on the front panel.

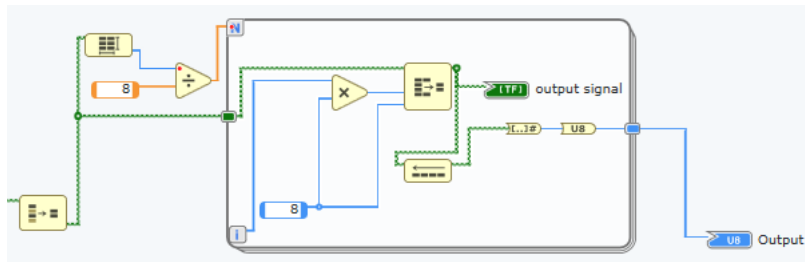


Figure 5: Phase 3 details showing Bit Reversal logic and U8 conversion at the receiver.

• Results and Analysis:

Figure 6 below displays the final system results verified on the LabVIEW NXG Front Panel using the input character 'A'. The original binary data is successfully converted into a Boolean array (01000001).

The "Manchester Encoded Signal" waveform chart strictly follows the specific logic defined for this project. The waveform starts with a High-to-Low transition, representing the first bit '0', followed by a Low-to-High transition, representing the

efficient graphical programming interface to visualize and analyze the system's performance.